

Anahita Bolourani

LinkedIn | Website | Github | Google Scholar | anabolourani@ucla.edu

Education

2024 – Present **University of California, Los Angeles (UCLA)**

Master of Science – Statistics and Data Science

2022 – 2026 **University of California, Los Angeles (UCLA)**

Doctor of Philosophy – Engineering

Publications

Banayeeanzade, A., Tak, A. N., Bahrani, F., Bolourani, A., Blas, L., Ferrara, E., Gratch, J., & Karimireddy, S. p. "Psychological Steering in LLMs: An Evaluation of Effectiveness and Trustworthiness", *arXiv preprint*, 2025.

Tak, A. N., Bolourani, A., Shank, D. B., & Gratch, J. "Impact of LLM Alignment on Impression Formation in Social Interactions", *COLM*, 2025.

Tak, A. N., Banayeeanzade, A., Bolourani, A., Kian, M., Jia, R., & Gratch, J. "Mechanistic Interpretability of Emotion Inference in Large Language Models," *Findings of ACL*, 2025.

Bolourani, A. Burton, H. Bozorgnia, Y. "Evaluating Procedures for Estimating Vertical Seismic Load Effects Specified in the U.S. Building Code," *ASCE (JSE)*, 2024.

Bolourani, A. et al. "Structural health monitoring of using support vector machine and principal component analysis," *Structures*, 2021.

Tak, A. N., Bolourani, A. et al. "BIM-based 4D mobile crane simulation and onsite operation management," *Automation in Construction*, 2021.

Research Experience

2025 – Present **Researcher, TrustworthyAI Lab, UCLA**

- Advance TabSyn diffusion models for heterogeneous tabular data using reinforcement learning
- long-context tabular reasoning via compression

2024 – Present **Independent Researcher**

- how LLM alignment shapes social-impression formation and emotion inference; results at COLM 2025
- Mechanistic interpretability of emotion inference in LLMs—showed that affective processing localizes to mid-layer MHSA units via layer-wise probes and activation patching results at Findings of ACL 2025

2022 – 2025 **Graduate Research Assistant, B. John Garrick Institute for the Risk Sciences, UCLA**

- Built a Python platform to stream/analyze Community Seismic Network and NASA JPL strong-motion data; extracted dominant structural frequencies (FFT/PSD) for real-time tracking

2019 – 2021 **Research Assistant, University of Tehran**

- Developed an SVM-PCA pipeline for structural-health monitoring
- Applied Self-Organizing Maps for unsupervised anomaly detection on benchmark structures under noise

Technical Skills

Programming: Python (Pandas, NumPy, PyTorch, SciPy, scikit-learn, ObsPy, Matplotlib), R

ML Tooling: Large Language Models (LLM), Diffusion Models, Machine Learning (ML)

Data/Dev Ops: Git, LaTeX, Docker

Selected Coursework

STATS M231A & STAT C261 Pattern Recognition & Machine Learning,
STATS 200A Applied probability,
STATS 200B Theoretical Statistics,
STATS 201A Research Design & Sampling,
STATS 202A Statistical Programming
STATS C216 Applied Bayesian Social Statistics,
STATS 213 Synthetic Data Generation,
STAT 218 Statistical Analysis of Networks

Certificates

“Statistical Learning,” Stanford Online
"Generative AI with Large Language Models", AWS
“Neural Networks and Deep Learning,” DeepLearning.AI
Google Data Analytics Professional Certificate

Awards & Honors

ASCE Metropolitan Los Angeles *Le Val Lund Memorial Award*, 2024
ASCE Los Angeles Section *Outstanding Graduate Student Scholarship*, 2023 & 2024
UCLA *Summer Mentored Research Fellowship*, 2022
UCLA *Graduate Student Researcher Award*, 2022